

Q1. A park ranger is tracking the population of deer in a forest.

The population P after t years is given by $P = \frac{100\sqrt{t+1}}{3}$.

Q2.

$$\frac{100\sqrt{44}}{3}$$

Q2.

$$\frac{100\sqrt{33}}{3}$$

Q2.

$$\frac{100\sqrt{42}}{3}$$

Q2.

$$\frac{1002\sqrt{4}}{3}$$

Q2.

$$\frac{100\sqrt{34}}{3}$$

A hiker is calculating her fuel consumption F after driving d miles. The consumption is given by $F = \frac{25\sqrt{d+5}}{11}$.

$$\frac{25\sqrt{1616}}{11}$$

$$\frac{25\sqrt{164}}{11}$$

$$\frac{254\sqrt{16}}{11}$$

$$\frac{25\sqrt{114}}{11}$$

$$\frac{25\sqrt{16}}{11}$$

The altitude A of a plane after h hours is given by $A = \frac{3000\sqrt{2h+1}}{7}$.

$$\frac{3000\sqrt{1515}}{7}$$

$$\frac{3000\sqrt{15}}{7}$$

$$\frac{3000\sqrt{153}}{7}$$

$$\frac{3000\sqrt{155}}{7}$$

$$\frac{300015}{7}$$

A wildlife expert is studying the growth of a bear population P after t years, given by $P = \frac{5\sqrt{3t+2}}{4}$.

$$\frac{5\sqrt{1414}}{4}$$

$$\frac{502\sqrt{14}}{4}$$

$$\frac{50\sqrt{14}}{4}$$

$$\frac{5014}{4}$$

$$\frac{50\sqrt{142}}{4}$$

A natural gas pipeline's flow rate F after d days is given by $F = \frac{1000\sqrt{d+10}}{15}$.

$$\frac{10005\sqrt{25}}{15}$$

$$\frac{1000\sqrt{25}}{15}$$

$$\frac{1000\sqrt{2525}}{15}$$

$$\frac{100025}{15}$$

$$\frac{1000\sqrt{255}}{15}$$

Challenge (Habanero)

Open Ended Questions (FRQ)

Q9. A forest's tree density D after t years is given by $D = \frac{200\sqrt{2t+5}}{t+5}$. Rationalize the denominator and simplify for $t = 9$. Show all steps.

Q10. The water level W in a lake after d days is given by $W = \frac{5\sqrt{d+1}}{d+1}$. Rationalize the denominator and simplify for $d = 8$. Explain your reasoning.

Chef's Tips

- Encourage students to use mathematical notation and provide clear explanations.
- Remind students to check their units and ensure they are consistent throughout their calculations.
- Suggest that students use real-world examples to help illustrate the concepts of rationalizing denominators.